ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $5n$	$5(n+3)-15=5n$
2	**(d)** $3n+3$	$n+(n+1)+\cdots=3n+3$
3	**(d)** 20	Top $=a+3b+3c+d$
4	**(b)** 90	Sum $=9\times$ middle date
5	**(c)** 1	$n^2-(n-1)(n+1)=1$
6	**(c)** $4n$	Expand: $4an=4\times 1n$
7	**(d)** $4n$	$4(n+6)-24=4n$
8	**(b)** $4n+6$	$n+(n+1)+\cdots=4n+6$
9	**(d)** 23	Top $=a+3b+3c+d$
10	**(b)** 108	Sum $=9\times$ middle date

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $=16+61=77$ (1) $=11\times 7$ (1).	1+1
12	$3(n+4)-12=3n$ (1); $\div 3$ gives n (1).	1+1
13	Numbers $2n+1$ and $2n+3$ (1); sum $=4n+4=4(n+1)$ (1).	1+1
14	Sum $=38+83=121$ (1) $=11\times 11$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Row 2: 5, 4, 5 (1); row 3: 9, 9 (1); top $=18$ (1).	1+1+1
16	$83-38=45$ (2) $=9\times 5$ (1).	2+1
17	Dates: 3, 4, 5, 10, 11, 12, 17, 18, 19 (1); sum $=99$ (1); $=9\times 11$ (1).	1+1+1
	OR: Row 2: 7, 5, 4 (1); row 3: 12, 9 (1); top $=21$ (1).	1+1+1
18	$94-49=45$ (2) $=9\times 5$ (1).	2+1
	OR: Dates: 5, 6, 7, 12, 13, 14, 19, 20, 21 (1); sum $=117$ (1); $=9\times 13$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Terms $n, n+1, \ldots, n+2$ (1); sum $3n+3$ (2); for $n=14$ the numbers add to 45 and $3\times 14+3=45$ (2).	1+2+2
	OR: $(2n+10)/2-n=n+5-n=5$ (3); for $n=7$: $(2\times 7+10)/2-7=5$ (2).	3+2
20	Terms $n, n+1, \ldots, n+3$ (1); sum $4n+6$ (2); for $n=9$ the numbers add to 42 and $4\times 9+6=42$ (2).	1+2+2
	OR: $(2n+14)/2-n=n+7-n=7$ (3); for $n=9$: $(2\times 9+14)/2-9=7$ (2).	3+2

Mistakes to avoid:

- Adding unlike terms: $3x+4x^2=7x^3$; only like terms are combined, and $3x+4x=7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a+b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x-3)=-x+3$, not $-x-3$.
- Writing $(a+b)^2=a^2+b^2$ or $(a-b)^2=a^2-b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n+1$ for $n=1,2,3$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1	$n^2-(n-1)(n+1)=1$
2	**(c)** $8n$	Expand: $4an=4\times 2n$
3	**(d)** $2n$	$2(n+8)-16=2n$
4	**(b)** $5n+10$	$n+(n+1)+\cdots=5n+10$
5	**(b)** 19	Top $=a+3b+3c+d$
6	**(a)** 144	Sum $=9\times$ middle date
7	**(c)** 1	$n^2-(n-1)(n+1)=1$
8	**(c)** $12n$	Expand: $4an=4\times 3n$
9	**(a)** $3n$	$3(n+9)-27=3n$
10	**(a)** $6n+15$	$n+(n+1)+\cdots=6n+15$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5(n+7)-35=5n$ (1); $\div 5$ gives n (1).	1+1
12	$n=12$: $5\times 12+10$ (1) $=70$ (1).	1+1
13	Sum $=49+94=143$ (1) $=11\times 13$ (1).	1+1
14	$2(n+9)-18=2n$ (1); $\div 2$ gives n (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Row 2: 7, 8, 5 (1); row 3: 15, 13 (1); top $=28$ (1).	1+1+1
16	$72-27=45$ (2) $=9\times 5$ (1).	2+1
17	Dates: 9, 10, 11, 16, 17, 18, 23, 24, 25 (1); sum $=153$ (1); $=9\times 17$ (1).	1+1+1
	OR: Row 2: 8, 5, 3 (1); row 3: 13, 8 (1); top $=21$ (1).	1+1+1
18	$61-16=45$ (2) $=9\times 5$ (1).	2+1
	OR: Dates: 12, 13, 14, 19, 20, 21, 26, 27, 28 (1); sum $=180$ (1); $=9\times 20$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Terms $n, n+1, \ldots, n+4$ (1); sum $5n+10$ (2); for $n=21$ the numbers add to 115 and $5\times 21+10=115$ (2).	1+2+2
	OR: $(2n+8)/2-n=n+4-n=4$ (3); for $n=12$: $(2\times 12+8)/2-12=4$ (2).	3+2
20	Terms $n, n+1, \ldots, n+5$ (1); sum $6n+15$ (2); for $n=7$ the numbers add to 57 and $6\times 7+15=57$ (2).	1+2+2
	OR: $(2n+16)/2-n=n+8-n=8$ (3); for $n=5$: $(2\times 5+16)/2-5=8$ (2).	3+2

Mistakes to avoid:

- Adding unlike terms: $3x+4x^2=7x^3$; only like terms are combined, and $3x+4x=7x$ (not $7x^2$).
- Writing the two-digit number as ab (product) instead of $10a+b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x-3)=-x+3$, not $-x-3$.
- Writing $(a+b)^2=a^2+b^2$ or $(a-b)^2=a^2-b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n+1$ for $n=1,2,3$).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Algebra Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 23	Top $= a + 3b + 3c + d$
2	**(a)** 171	Sum $= 9\times$ middle date
3	**(b)** 1	$n^2 - (n-1)(n+1) = 1$
4	**(c)** $16n$	Expand: $4an = 4 \times 4n$
5	**(c)** $6n$	$6(n+5) - 30 = 6n$
6	**(c)** $7n + 21$	$n + (n+1) + \cdots = 7n + 21$
7	**(c)** 25	Top $= a + 3b + 3c + d$
8	**(a)** 207	Sum $= 9\times$ middle date
9	**(c)** 1	$n^2 - (n-1)(n+1) = 1$
10	**(d)** $20n$	Expand: $4an = 4 \times 5n$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$n = 20$: $6 \times 20 + 15$ (1) $= 135$ (1).	1+1
12	Sum $= 27 + 72 = 99$ (1) $= 11 \times 9$ (1).	1+1
13	$6(n+6) - 36 = 6n$ (1); $\div 6$ gives n (1).	1+1
14	$n = 33$: $4 \times 33 + 6$ (1) $= 138$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Row 2: 4, 6, 7 (1); row 3: 10, 13 (1); top $= 23$ (1).	1+1+1
16	$95 - 59 = 36$ (2) $= 9 \times 4$ (1).	2+1
17	Dates: 14, 15, 16, 21, 22, 23, 28, 29, 30 (1); sum $= 198$ (1); $= 9 \times 22$ (1).	1+1+1
	OR: Row 2: 8, 4, 5 (1); row 3: 12, 9 (1); top $= 21$ (1).	1+1+1
18	$82 - 28 = 54$ (2) $= 9 \times 6$ (1).	2+1
	OR: Dates: 16, 17, 18, 23, 24, 25, 30, 31, 32 (1); sum $= 216$ (1); $= 9 \times 24$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Terms $n, n+1, \ldots, n+6$ (1); sum $7n + 21$ (2); for $n = 15$ the numbers add to 126 and $7 \times 15 + 21 = 126$ (2).	1+2+2
	OR: $(2n + 12)/2 - n = n + 6 - n = 6$ (3); for $n = 20$: $(2 \times 20 + 12)/2 - 20 = 6$ (2).	3+2
20	Terms $n, n+1, \ldots, n+7$ (1); sum $8n + 28$ (2); for $n = 30$ the numbers add to 268 and $8 \times 30 + 28 = 268$ (2).	1+2+2
	OR: $(2n + 18)/2 - n = n + 9 - n = 9$ (3); for $n = 6$: $(2 \times 6 + 18)/2 - 6 = 9$ (2).	3+2

Mistakes to avoid:

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- Writing the two-digit number as ab (product) instead of $10a + b$.
- Checking a trick or identity with one example only and calling it a proof; use a variable for the general case.
- Sign mistakes when removing brackets: $-(x - 3) = -x + 3$, not $-x - 3$.
- Writing $(a + b)^2 = a^2 + b^2$ or $(a - b)^2 = a^2 - b^2$ while simplifying expressions.
- Deriving a pattern rule from the first term only; check that the rule works for every given term (e.g. $3n + 1$ for $n = 1, 2, 3$).

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